

1. Consider M/M/1: This is the simplest queueing system, where the arrival is considered Poisson Process and service mechanism, Exponential Process. The system consists of only one server, one unlimited FIFO queue and unlimited customer population. Derive the formulas for
 - a. Probability that there are n entities in the system
 - b. Average entities in the system (L_s)
 - c. Average entities in the queue (L_q)
 - d. Average Waiting time in the system (W_s)
 - e. Average Waiting time in the queue (W_q)
2. Calls arrive at a call booth at the rate 110 calls per second. The average time to respond to a call is 200 seconds (i.e. mean service time = 200 s).
 - a) Determine whether the system is stable (does a steady-state exist?) and show the computation.
 - b) If the system were stable, compute the steady-state probability that at least 5 calls are present in the system (waiting + in service). If it is unstable, explain what that implies for that probability.
3. Customers arrive at a help desk according to a Poisson process at 4 customers per minute. Each customer's service time is exponential with mean 1/6 minute (i.e. service rate $\mu=6$ per minute).
 - a) Compute the traffic intensity ρ .
 - b) Find the average waiting time in queue W_q and the average time in system W .
 - c) What is the probability that an arriving customer must wait?
 - d) What is the probability that a customer's waiting time in queue exceeds 1 minute? (Give the exact expression and numeric value.)
4. Jobs arrive at a server at rate $\lambda=2$ jobs/minute. Service times are exponential with mean 0.15 minute (so $\mu \approx 6.6667$ per minute).
 - a) Compute the steady-state probability $P\{N=n\}$ for an arbitrary n .
 - b) Using the expression in (a), compute the probability that there are at least 3 jobs in the system: $P\{N \geq 3\}$.
 - c) Compute the long-run fraction of time the server is idle.
5. A web server is modeled as M/M/1. The system should have on average no more than 2 requests in the system (waiting + service) to meet SLAs. Service rate is $\mu=20$ requests per minute.
 - a) Find the maximum arrival rate $\lambda_{\max}$ (in requests/minute) that satisfies $L \leq 2$.
 - b) If the arrival rate is set to $\lambda = 1/2 \lambda_{\max}$, compute the expected waiting time in queue and the probability that an arrival finds the server busy.